

Highlights

Sparsity-Adaptive Gated Eviction for Memory-Efficient Long-Context Inference of Large Language Models

- KV cache eviction is unified as one global memory-contract allocation
- Layer-adaptive budgets re-allocate cache capacity online from attention entropy
- Decayed confidence with a protected reservoir keeps 94.8% retrieval at 128K
- A 20% cache budget retains 98.0% of the LongBench average across 11 tasks
- KV memory shrinks 4.4x and decode throughput reaches 2.02x at 128K context